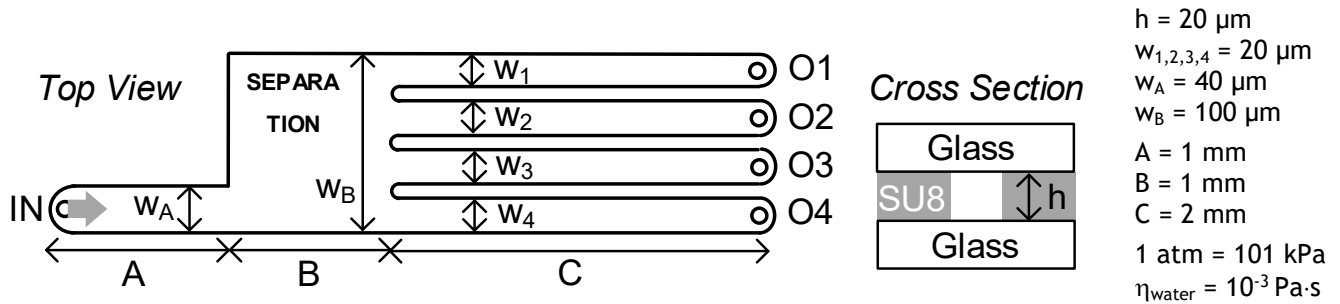


Politecnico di Milano
Biochip A.Y. 2021/2022 - M. Carminati
 Exam of 19.07.2023

*Test duration: 2 hours - No support material (textbooks, laptops, smartphones, notes) admitted.
 All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.*

Exercise 1

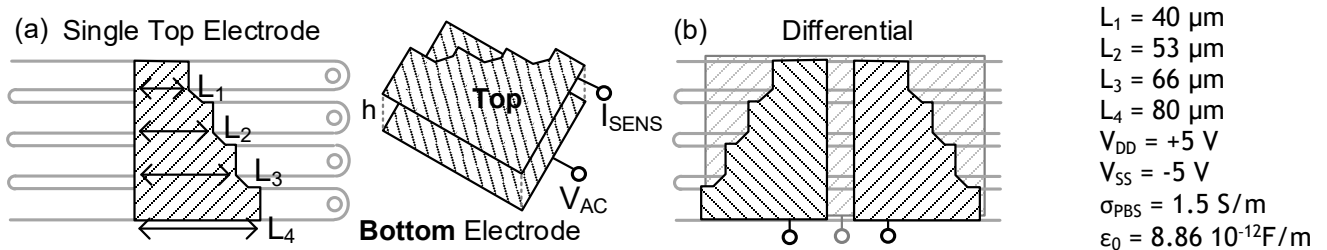
Consider the following microfluidic device for separation and electrical counting of bioparticles. Mixed populations of particles enter from single inlet IN. In section B particles are separated by means of transversal electric fields. Laterally separated particles are then counted in 4 output sections C. Electrodes (gold) are patterned on both glass slides forming the bottom and top lid of the sandwich device separated by SU8 in which channels are etched.



- Draw and compute the electrical equivalent of the fluidic circuit assuming it filled with water.
- Assuming $P_{IN} = 6$ atm and open outlets, find the water speed in every section.
- If a syringe pump with a 10mL syringe is used in the conditions of point (b) what would be the operating time?
- How would results of point (b) change in case of an increase of temperature?
- Considering the need to create a transversal force acting on particles, compare *electrophoresis* and *dielectrophoresis* (for instance in terms of particle types, electrodes geometries, driving signals etc...).
- Describe the fabrication steps of the device, including gold electrodes on glass and the microchannels.

Exercise 2

Now focus on the impedance detection section based on **parallel-plate electrodes**: in order to save space and cost, a **single TIA** is used connected to a common trapezoidal sensing electrode covering all channels. Thanks to the different longitudinal length (L_1 to L_4) in each output channel, the channel number where each particle is flowing is identified from the temporal length of the resistive pulse.



- Assuming the channels filled with PBS, find the electrical impedance between the bottom (forcing) and top (sensing) electrode. Draw the Bode plot of the impedance spectrum. Identify the optimal sensing frequency (*suggestion* consider individual impedance sections in parallel).
- Draw the scheme of the impedance readout circuit and size its components (*suggestion* start from the requirement of the sensing frequency). Should a second gain stage be added after the TIA?
- Draw the signal at the output vs time of the LIA for a particle of 10 μm diameter flowing in section 1, 2, 3 and 4. Assuming a particle speed of 1 μm/ms, set the proper LIA bandwidth.
- Study the noise of the proposed circuit, considering a voltage noise of 10 nV/sqrt(Hz) for all opamps.
- Now consider a differential readout (Fig. b). Discuss the advantages of differential sensing and draw the scheme of a differential readout circuit.
- Repeat point (c) in the case of the differential readout (draw the signal vs time at the output of the LIA).

Multichance students with 30% reduction of the exercises please skip points e) and f) of both exercises.